De novo bispecific binders for Alzheimer's disease

Aβ42 recognition × TfR1 targeting | Goal: receptor-mediated brain delivery

01 TARGETS

Aβ42

TfR1

02 ARM DESIGN

62 accepted arms (reported)

380

BindCraft · ProteinMPNN

1,342 / 2,051 trajectories

5 + 5

03 FUSION LIBRARY

250

5 × 5 arm pairs

5 linkers × 2 domain orders

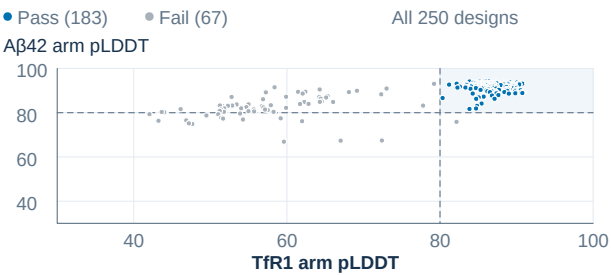
04 FOLD SCREEN

183 of 250

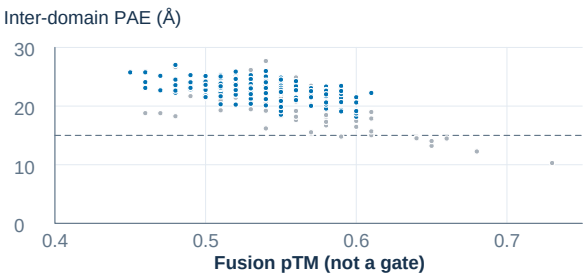
73.2% pass Stage 8

AlphaFold2 / ColabFold

a Per-arm folding confidence



b Whole-fusion confidence and uncertainty



Gates: both arms pLDDT >80; inter-domain PAE >15 Å. High PAE = uncertain relative placement.

Computational fold screen; dual-target binding and BBB transport remain untested. Molecular drawings are schematic.